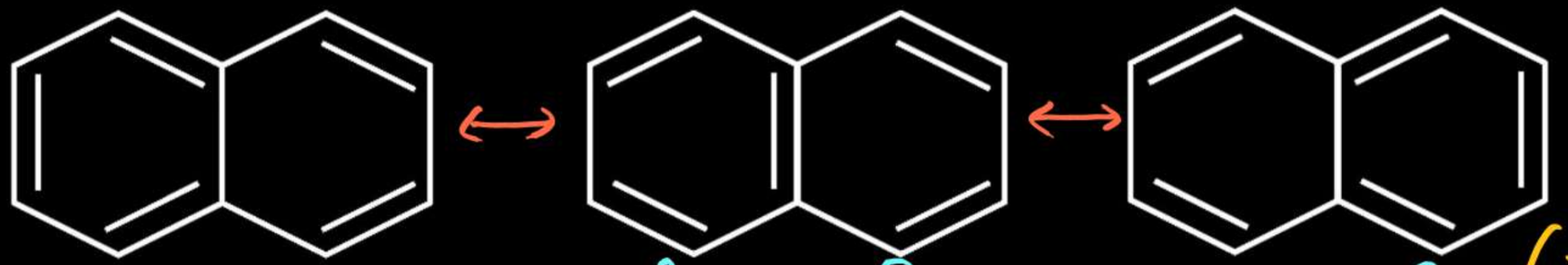


GOC

SKM SIR (SURENDRA MISHRA SIR)

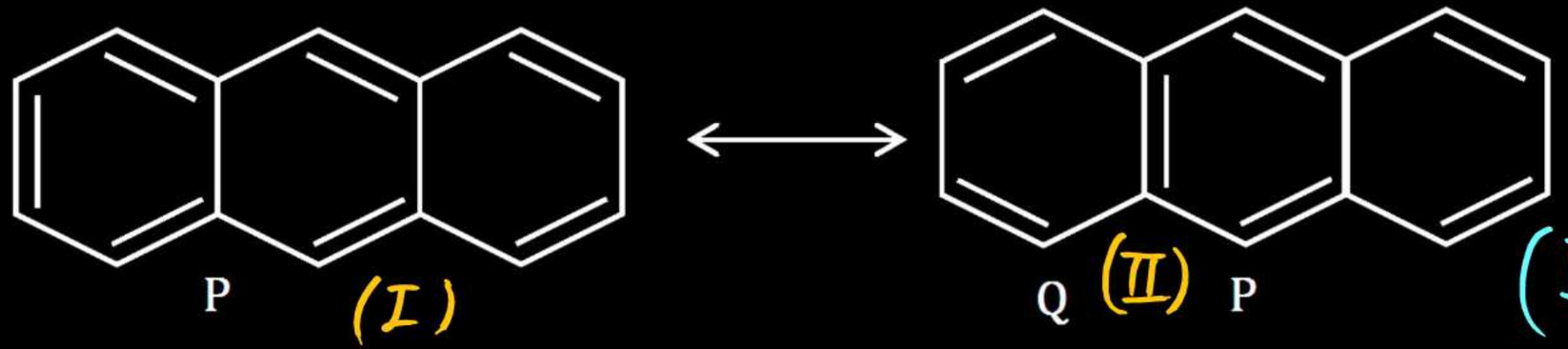
Genral Organic Chemistry

(10)

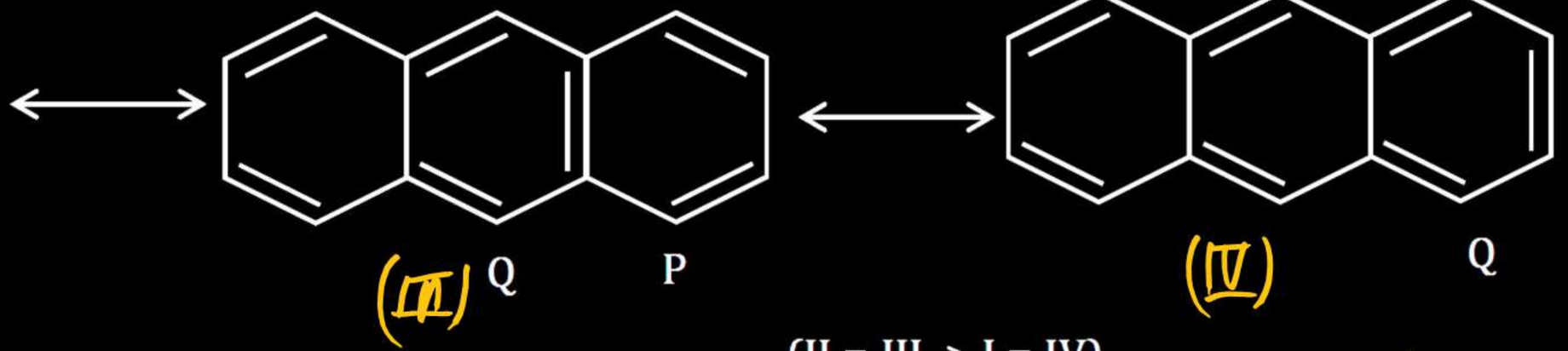


(II > I = III)

(11)



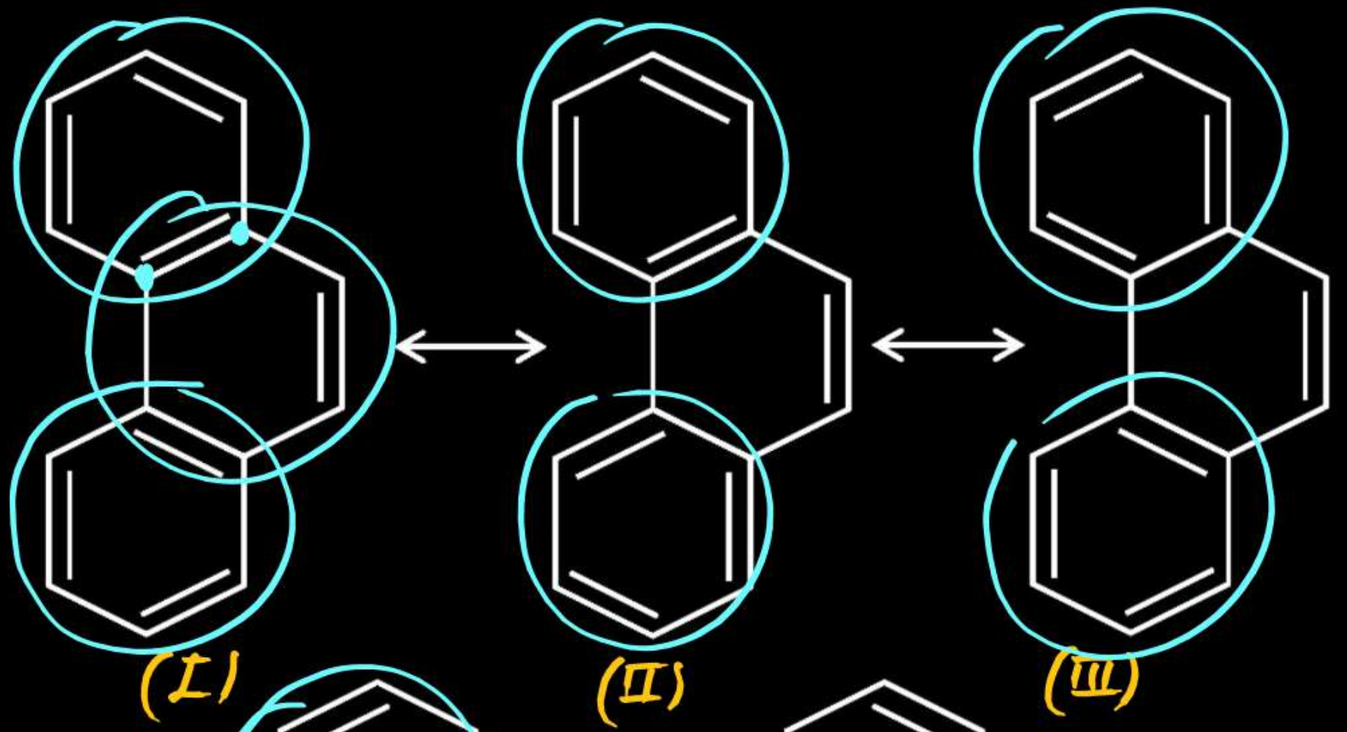
(III = II > I = IV)

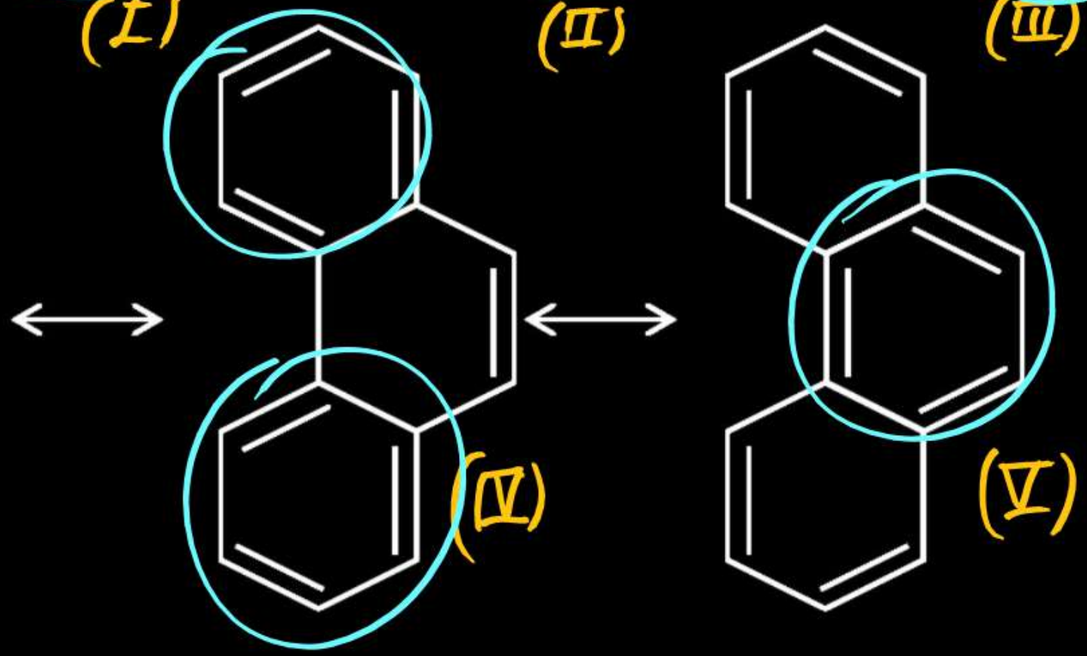


(II = III > I = IV)

Genral Organic Chemistry

(12)



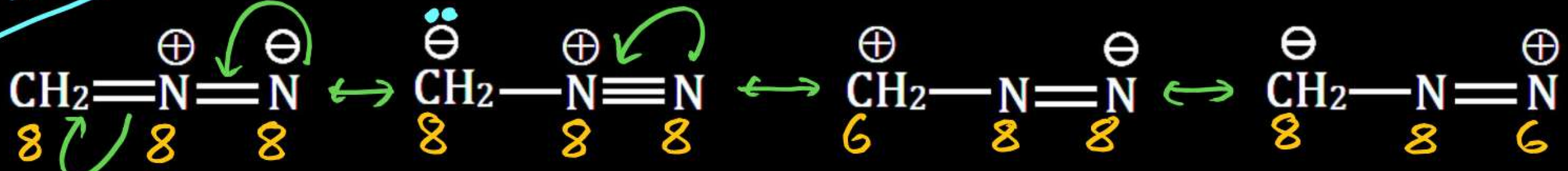


I > II = III = IV > V

Genral Organic Chemistry

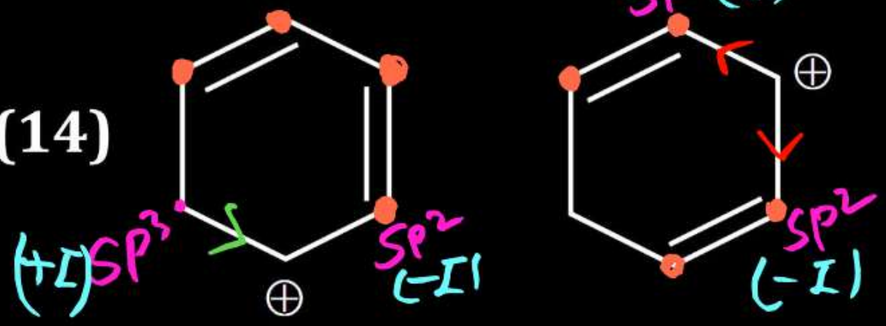
JEE ADV PYQ.

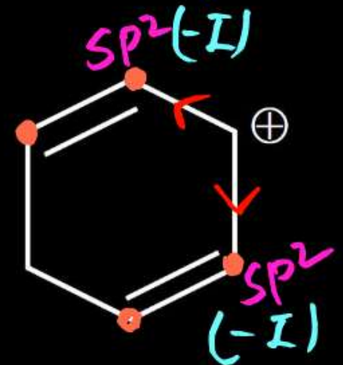
(13)



I > II > III > IV

(14)



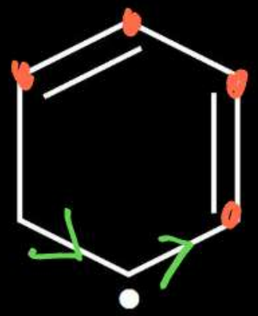


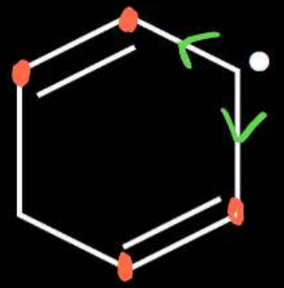
(I > II)

$-\overset{SP^2}{C}H=CH_2$ (-I) EWG

$-\overset{SP^3}{C}H_2-CH_3$ (+I) EDG

(15)

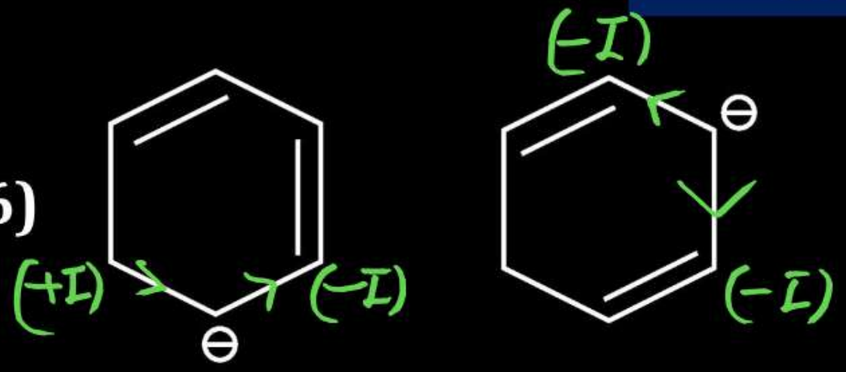




(I > II)

Genral Organic Chemistry

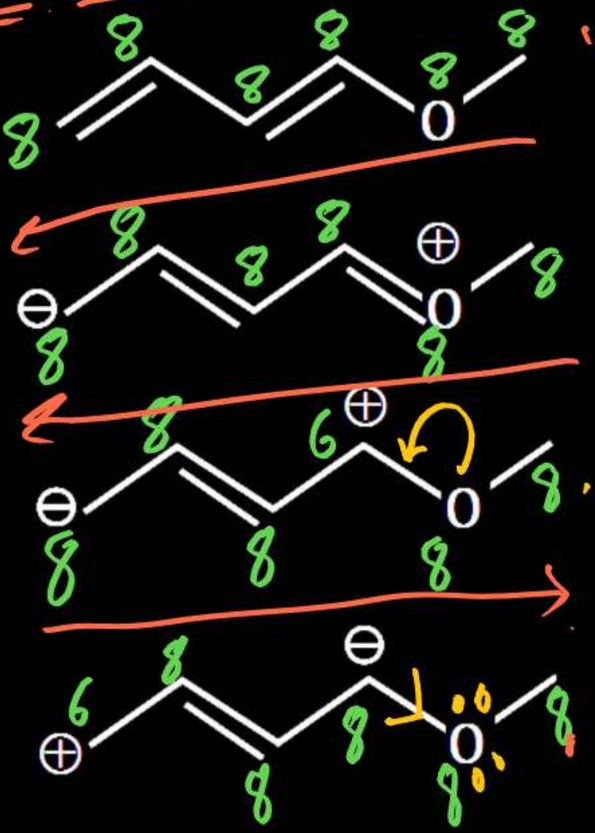
(16)



(II > I)

JEE Adv PYQ:

(17)



1 > 2 > 3 > 4

(I) $CH_2=CH_2$ — No Resonance

(II) $CH_2=CH-O-CH_3$ — Resonance phenomenon

$\Rightarrow$ π electron density (II > I)

(II) $CH_2=CH-O-CH_3$

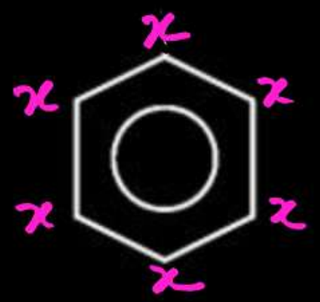
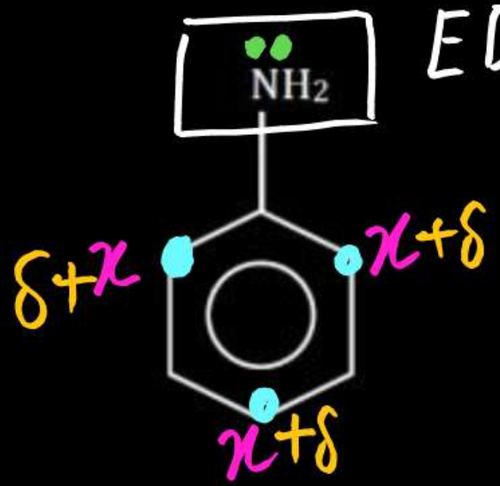
- F
- Cl
- Br
- I

- OR
- NH$_2$
- NHR
- NR$_2$

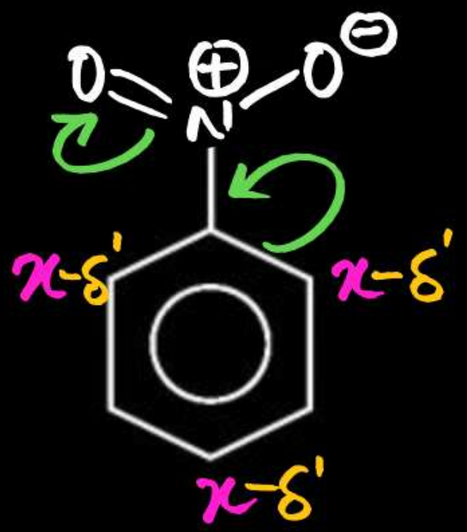
- SH
- SO_3H
- $O^{\ominus}$
- $NH^{\ominus}$
- $\overset{\ominus}{C}H_2$

Genral Organic Chemistry

Resonance / mesomeric Effect :-

	Resonance	Resonance Effect
	✓	X
	✓	✓

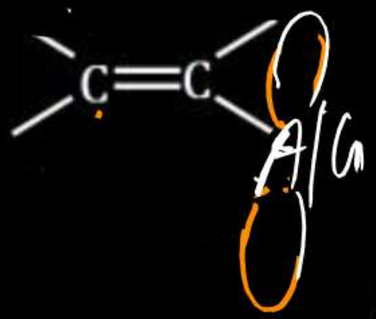
Genral Organic Chemistry



✓ ✓

Resonance/ Mesomeric Effect (R/M effect):-

Permanent displacement of π e– density due to P orbital of directly attached atom/group is known as R/M effect of that atom/group on conjugated system.

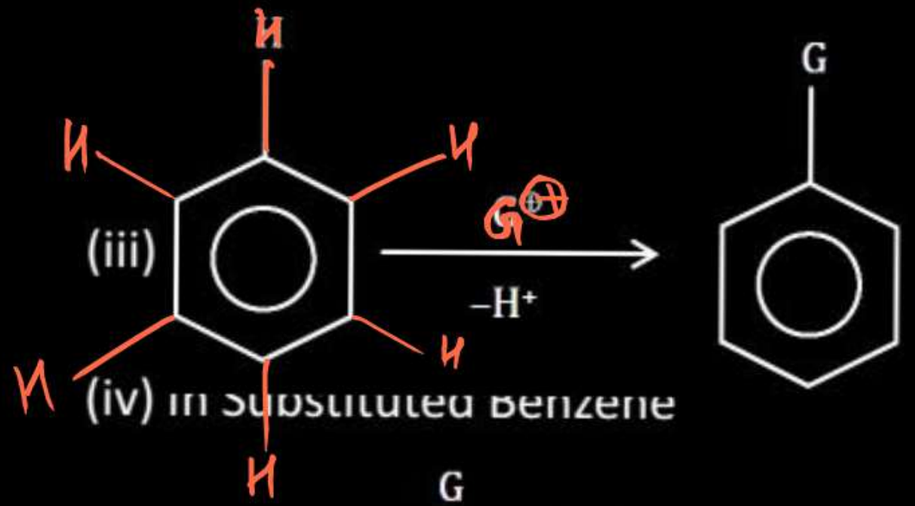


"P" orbital

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Note:- (i) Permanent effect

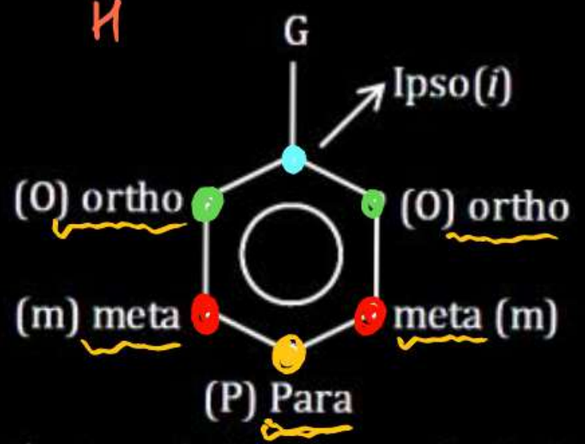
(ii) Distance independent effect



(iii)

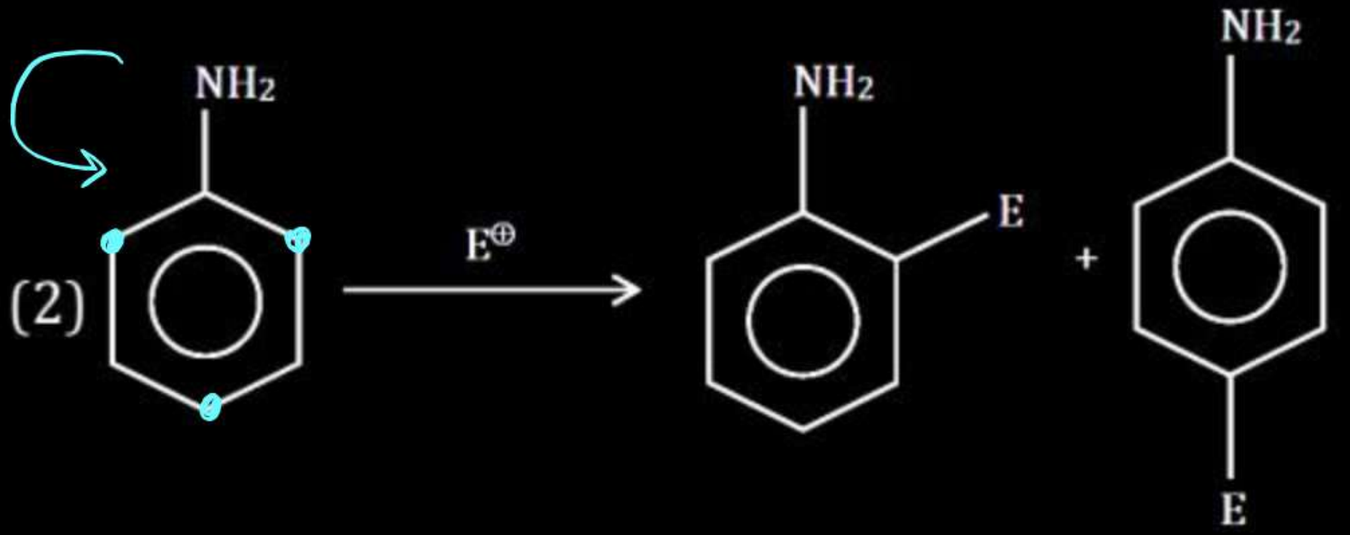
(Electrophilic Substitution Reaction)

(iv) In Substituted Benzene

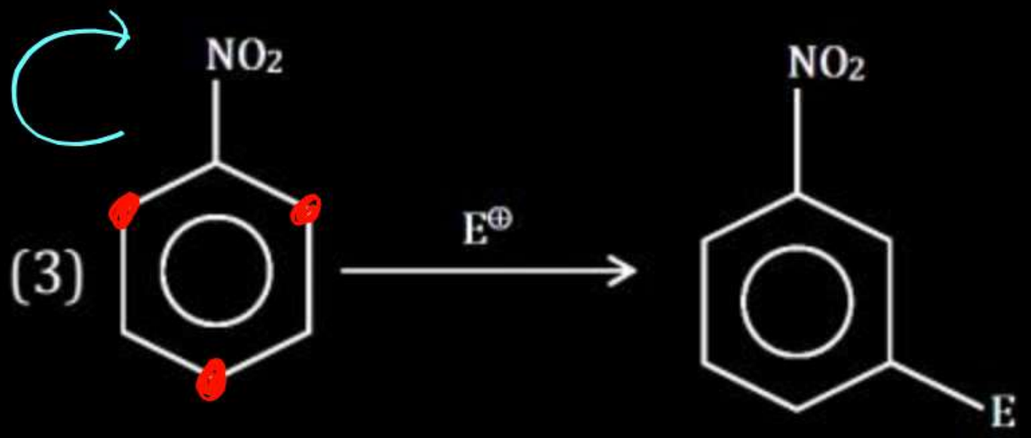


(v) Electrophilic Aromatic Substitution

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[ortho & para Substitution product]



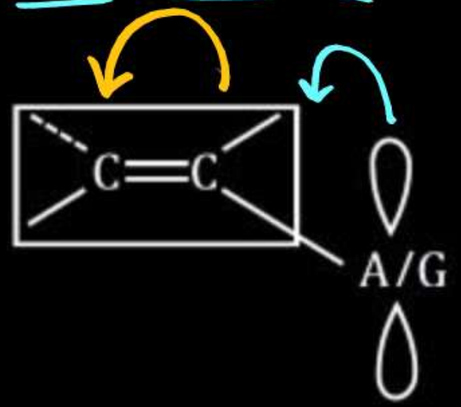
(meta product)

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Type of Resonance effect:-

⇒ There are two type of Resonate effect

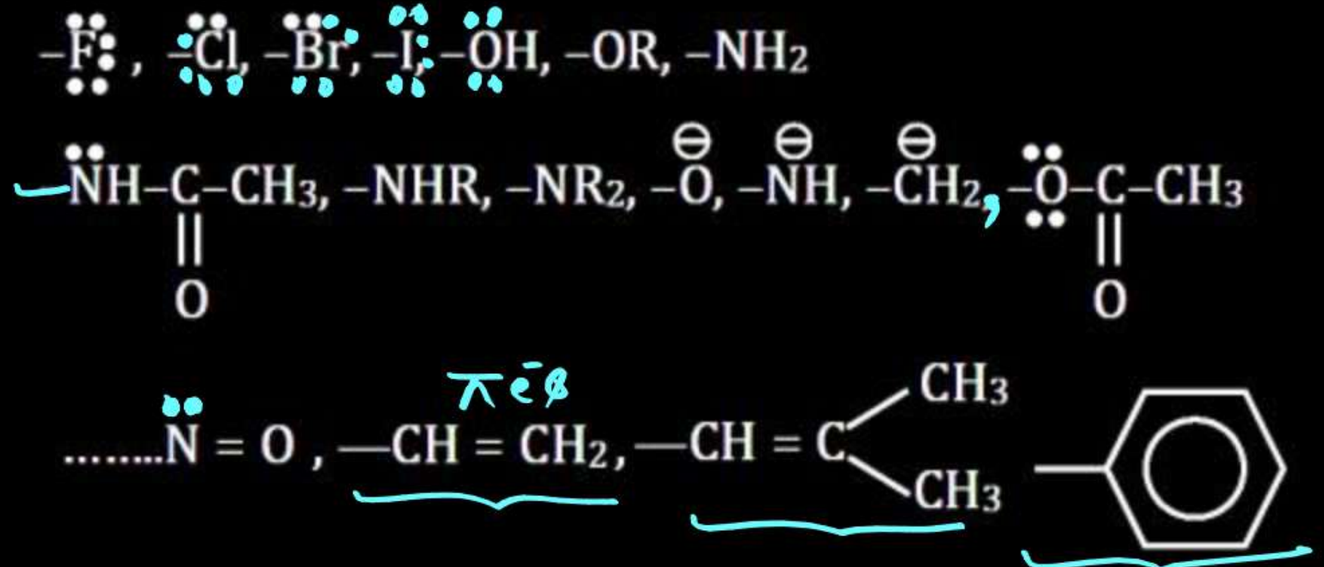
(i) + Resonance/+ mesomeric effect (+R / +M Effect): permanent polarisation of πe⁻ density due to "P" orbital of attached atom/group away to itself is known as +R/+M effect of A/G "P" orbital



Note :- (i) +R/+M Effect is Electron donating Effect

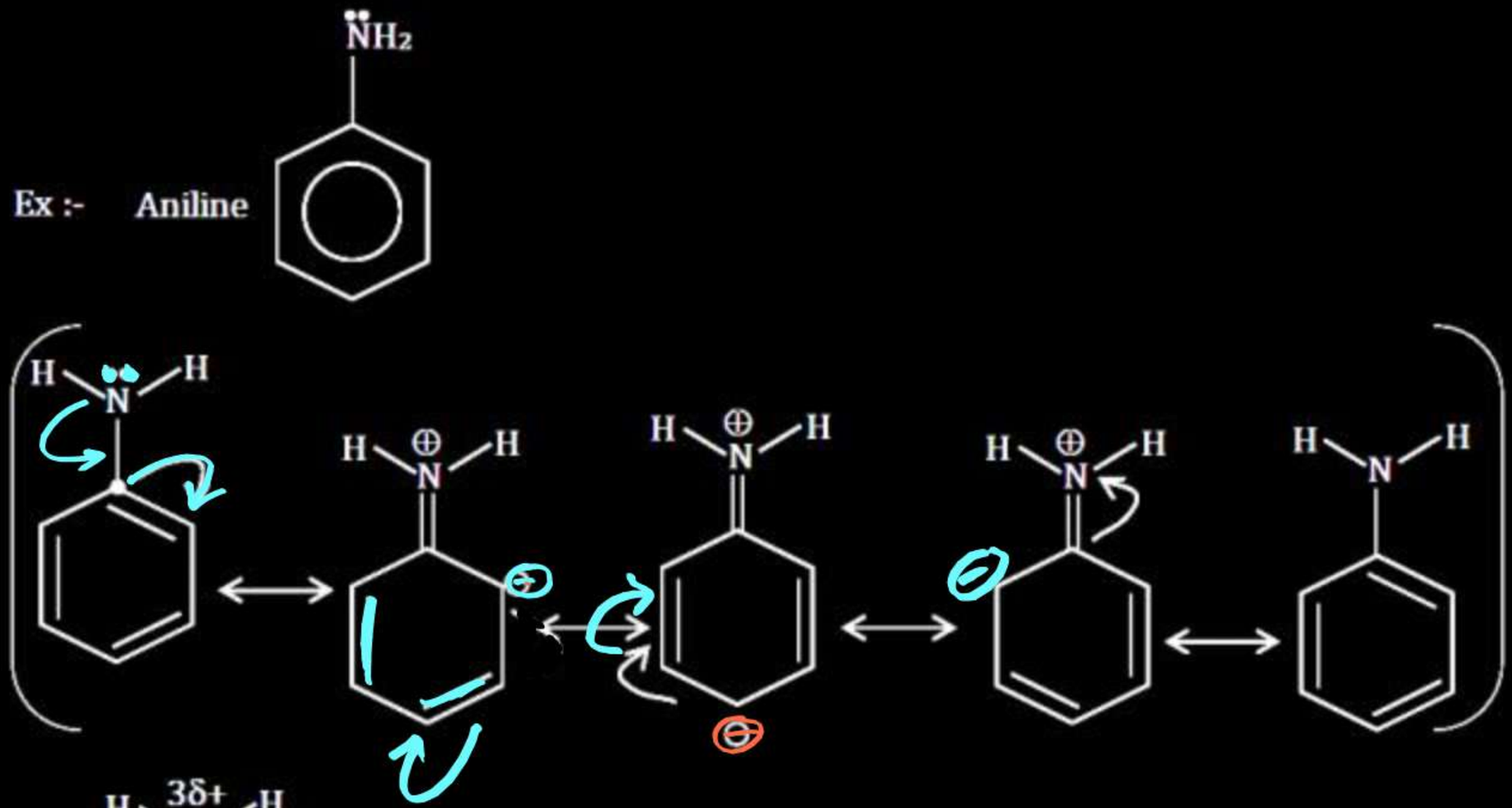
(ii) A/G which can show +R effect / +M effect

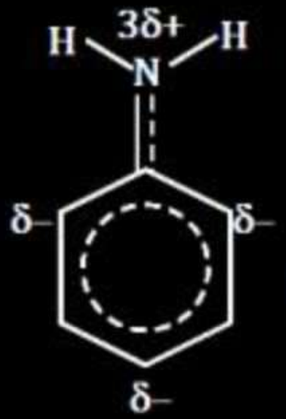
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Ex :- Aniline





(Resonance effect)

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Ex :- Phenoxide Ion

